



Python Control Structures

Conditional Statements

1. If Statement

Explanation: Check if a number is positive.

```
num = 5
if num > 0:
    print("Positive number") # Runs if num is positive
```

Output:

Positive number

Homework:

Check if number is negative or zero.

2. If-Else Statement

Explanation: Check if a number is even or odd.

```
num = 4
if num % 2 == 0:
    print("Even number")
else:
    print("Odd number")
```

Output:

Even number

Homework:

Ask user input and check even or odd.



3. If-Elif-Else Statement

Explanation: Assign grade based on score.

```
score = 85
if score >= 90:
    print("Grade A")
elif score >= 80:
    print("Grade B")
else:
    print("Grade C")
```

Output:

Grade B

Homework:

Add input for score and handle invalid values.

4. Nested If

Explanation: Check if number is positive and even.

```
num = 8
if num > 0:
    if num % 2 == 0:
        print("Positive even")
    else:
        print("Positive odd")
```

Output:

Positive even

Homework:

Check for negative even and negative odd.



5. Logical Operators

Explanation: Use and to check range.

```
num = 15
if num > 10 and num < 20:
    print("Between 10 and 20")
```

Output:

Between 10 and 20

Homework:

Use or to check if number is less than 5 or greater than 15.

For Loops

6. Range with 1 Parameter

Explanation: Loop 0 to 4.

```
for i in range(5):
    print(i)
```

Output:

0
1
2
3
4

Homework:

Print all odd numbers from 0 to 9.



7. Range with 2 Parameters

Explanation: Loop 1 to 5.

```
for i in range(1, 6):  
    print(i)
```

Output:

1
2
3
4
5

Homework:

Print numbers from 10 to 1 in reverse.

8. Range with 3 Parameters

Explanation: Loop with step 2.

```
for i in range(0, 10, 2):  
    print(i)
```

Output:

0
2
4
6
8

Homework:

Print all even numbers between 20 to 30.



9. For with List

Explanation: Loop through items in list.

```
fruits = ["apple", "banana", "cherry"]  
for fruit in fruits:  
    print(fruit)
```

Output:

apple
banana
cherry

Homework:

Create a list of 5 countries and print them.

10. For with String

Explanation: Loop through characters in string.

```
word = "hello"  
for char in word:  
    print(char)
```

Output:

h
e
l
l
o

Homework: Print only vowels from a given word.



11. For with Break

Explanation: Stop loop early.

```
for i in range(5):  
    if i == 3:  
        break  
    print(i)
```

Output:

0
1
2

Homework: Break the loop when number is divisible by 4.

12. For with Continue

Explanation: Skip an iteration.

```
for i in range(5):  
    if i == 2:  
        continue  
    print(i)
```

Output:

0
1
3
4

Homework:

Skip all even numbers from 0 to 9.



13. For with Pass

Explanation: Use pass as placeholder.

```
for i in range(3):  
    if i == 1:  
        pass  
    print(i)
```

Output:

0
1
2

Homework: Use pass inside if-else and print message outside.

14. Reverse For Loop

Explanation: Loop backward.

```
for i in range(5, 0, -1):  
    print(i)
```

Output:

5
4
3
2
1

Homework:

Reverse print from 10 to 1 skipping 2 numbers each time.



15. For-Else Loop

Explanation: Else after loop.

```
for i in range(3):  
    print(i)  
else:  
    print("Done")
```

Output:

0
1
2
Done

Homework: Use break so else doesn't run.

While Loops

16. Basic While

Explanation: Loop while condition true.

```
i = 0  
while i < 3:  
    print(i)  
    i += 1
```

Output:

0
1
2

Homework: Print table of 5 using while loop.



17. While with Break

Explanation: Stop loop using break.

```
count = 0
while True:
    print(count)
    if count == 2:
        break
    count += 1
```

Output:

0
1
2

Homework: Break when number is divisible by 7.

18. While with Continue

Explanation: Skip using continue.

```
i = 0
while i < 5:
    i += 1
    if i == 3:
        continue
    print(i)
```

Output:

1
2
4
5

Homework: Skip all numbers divisible by 3 between 1 to 10.



19. While with Else

Explanation: Use else with while.

```
i = 0
while i < 3:
    print(i)
    i += 1
else:
    print("Loop ended")
```

Output:

0
1
2
Loop ended

Homework: Add break before else to stop it.

20. Infinite While

Explanation: Loop forever (use with caution).

```
# while True:
#     print("Running")
```

Output:

(Runs forever if uncommented)

Homework: Add counter and stop after 5 prints.



Special Keywords

21. Use of Break

Explanation: Exit early from loop.

```
for i in range(5):  
    if i == 2:  
        break  
    print(i)
```

Output:

0
1

Homework: Find first number divisible by both 3 and 5 from 1 to 50.

22. Use of Continue

Explanation: Skip current iteration.

```
for i in range(5):  
    if i == 3:  
        continue  
    print(i)
```

Output:

0
1
2
4

Homework:

Print all numbers from 1 to 20 except multiples of 3.



23. Use of Pass

Explanation: Do nothing.

```
for i in range(3):  
    if i == 1:  
        pass  
    print(i)
```

Output:

0
1
2

Homework: Use pass with multiple conditions.

24. Nested If-Else in Loop

Explanation: Mix control structures.

```
for i in range(5):  
    if i % 2 == 0:  
        print(i, "Even")  
    else:  
        print(i, "Odd")
```

Output:

0 Even
1 Odd
2 Even
3 Odd
4 Even

Homework: Add check to print "small even" if number is even and <3.



25. Sum of Range

Explanation: Sum numbers in a range.

```
total = 0
for i in range(1, 6):
    total += i
print(total)
```

Output:

15

Homework:

Print the sum of all even numbers from 1 to 50.